Hades Model Report

v0.8.4 GitHub

Hastie & Smith, et al. (2026)

MODEL CONFIGURATION

This report summarises the calibration of the model against the reference dataset, including parameter estimates and convergence diagnostics.

Mode	**Subalkaline**
Submode	**Tholeiitic**
Data suite	**./thingmuli_hekla.xlsx**
FeO/Fe^{2+}	**0.9**
Stages Run	Primitive Basic Intermediate Evolved

1. Results at a Glance

All of the mineral abundances and fractional crystallisation (FC) values for each phase are shown in the table below, as well as the resultant total fractions for the performed amount of fractional crystallisation over all phases.

The total melt fraction crystallised across all phases is 87.86 wt.%

	Primitive	Basic	Intermediate	Evolved	Total
Olivine	27.34	7.31		20.56	6.21
Clinopyroxene	1.89	37.72	19.45	30.93	34.23
Orthopyroxene			10.31		1.9
Plagioclase	56.31	42.39	56.27	1.05	44.17
K-feldspar				41.59	0.78
Quartz				0.02	0.0
Magnetite	10.16	7.09	8.83	0.16	7.28
Ilmenite	4.31	4.74	3.03	4.45	4.42
Apatite		0.75	2.0	0.75	0.98
Zircon			0.1	0.5	0.03
FC%	18.0	70.0	54.0	12.0	

Table 1: Optimised mineral weights for the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

2. Parameter Configuration

The parameters below define the model as it was executed for this run.

	Primary	Primitive	Basic	Intermediate	Evolved
Plagioclase	0.95	0.95	0.95	0.95	0.95
Clinopyroxene	0.25	0.25	0.25	0.25	0.25
Orthopyroxene	0.29	0.29	0.29	0.29	0.29
Amphibole	0.37	0.37	0.37	0.37	0.37
K-feldspar	1.3	1.3	1.3	1.3	1.3
Biotite	1.0	1.0	1.0	1.0	1.0
Titanaugite	1.0	1.0	1.0	1.0	1.0
Nepheline	1.0	1.0	1.0	1.0	1.0
Leucite	1.0	1.0	1.0	1.0	1.0
Nosean/Hauyne	1.0	1.0	1.0	1.0	1.0
Alkali-amphibole	1.0	1.0	1.0	1.0	1.0

Table 2: Partition coefficients for the model phases.

3. Magma Evolution

Initial Compositions

Primary	Primitive	Basic	Intermediate	Evolved	End
Sample	T3-01	T7-03	H4 11	0409-14	T3-12
SiO2	46.81	48.43	59.18	71.57	76.55
TiO2	2.65	2.65	1.3	0.33	0.14
Al2O3	14.79	14.03	15.04	14.2	12.64
Fe2O3	15.25	14.63	11.67	4.52	2.02
MgO	7.27	6.41	1.92	0.18	0.01
CaO	10.7	11.12	5.3	1.3	0.51
Na2O	2.3	2.44	4.17	5.42	4.73
K2O	0.23	0.29	1.42	2.47	3.41
P	1440.78	1659.08	2270.32	174.64	43.66
Ba	101.0	108.0	390.0	705.0	834.0
Rb	12.0	13.0	39.0	62.0	103.0
Sr	244.0	230.0	0.0	164.0	57.0
Nb	15.0	15.0	56.0	113.0	131.0
Zr	142.0	133.0	645.0	699.0	329.0
Y	29.0	29.0	0.0	179.0	194.0
La	12.0	16.0	55.0	99.0	136.0
Ce	30.0	33.0	117.0	180.0	243.0

	Primary	Primitive	Basic	Intermediate	Evolved	End
Nd		0.0	0.0	72.0	0.0	0.0
Sm		0.0	0.0	12.5	0.0	0.0
Eu		0.0	0.0	4.23	0.0	0.0
Gd		0.0	0.0	15.4	0.0	0.0
Tb		0.0	0.0	1.97	0.0	0.0
Dy		0.0	0.0	0.0	0.0	0.0
Ho		0.0	0.0	0.0	0.0	0.0
Er		0.0	0.0	0.0	0.0	0.0
Tm		0.0	0.0	1.06	0.0	0.0
Yb		0.0	0.0	7.1	0.0	0.0
Lu		0.0	0.0	1.0	0.0	0.0
Hf		0.0	0.0	13.6	0.0	0.0
Ta		0.0	0.0	3.74	0.0	0.0
Th		0.0	0.0	5.8	0.0	0.0
U		0.0	0.0	1.7	0.0	0.0

Table 3: Major oxide compositions of the model phases (wt.% for oxides, ppm for trace elements).

Computed Compositions

	Primary	Primitive	Basic	Intermediate	Evolved	End
SiO2		46.81	48.43	59.75	74.52	78.04
TiO2		2.65	2.65	1.43	0.59	0.34
Al2O3		14.79	14.03	15.27	15.19	16.01
Fe2O3		15.25	14.63	12.12	5.68	2.97
MgO		7.27	6.41	2.15	0.61	0.46
CaO		10.7	11.12	4.62	-1.3	-2.39
Na2O		2.3	2.44	4.55	5.82	6.11
K2O		0.23	0.29	0.89	1.73	1.78
P		1440.78	1659.08	2092.04	-64.86	-274.64
Ba		101.0	108.0	317.3	543.6	286.16
Rb		12.0	13.0	40.7	87.26	96.83
Sr		244.0	230.0	258.26	117.58	90.7
Nb		15.0	15.0	44.85	64.1	52.06
Zr		142.0	133.0	414.93	337.37	55.42
Y		29.0	29.0	54.72	53.7	50.1
La		12.0	16.0	26.13	33.33	34.99
Ce		30.0	33.0	46.38	44.1	44.94
Nd		0.0	0.0			
Sm		0.0	0.0			
Eu		0.0	0.0			
Gd		0.0	0.0			
Tb		0.0	0.0			
Dy		0.0	0.0			
Ho		0.0	0.0			
Er		0.0	0.0			
Tm		0.0	0.0			

	Primary	Primitive	Basic	Intermediate	Evolved	End
Yb		0.0	0.0			
Lu		0.0	0.0			
Hf		0.0	0.0			
Ta		0.0	0.0			
Th		0.0	0.0			
U		0.0	0.0			

Table 4: Major oxide compositions of the model phases (wt.%).

4. Optimisation Results

Primitive > Basic

PRIMITIVE

Population size	50
Iterations	500/500
Cost function	0.06
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	46.81	48.43	48.47	0.09
TiO2	2.65	2.65	2.65	0.04
Al2O3	14.79	14.03	14.03	0.0
Fe2O3	15.25	14.63	14.63	0.02
MgO	7.27	6.41	6.41	0.07
CaO	10.7	11.12	11.11	0.09
Na2O	2.3	2.44	2.44	0.0
K2O	0.23	0.29	0.26	9.2

Table 5: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	18.0
Olivine	27.34
Spinel	0.0
Clinopyroxene	1.89
Orthopyroxene	0.0
Plagioclase	56.31
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0

	Value
Magnetite	10.16
Ilmenite	4.31

Table 6: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	39.21	2.69	18.22	18.1	11.17	8.84	1.67	0.1

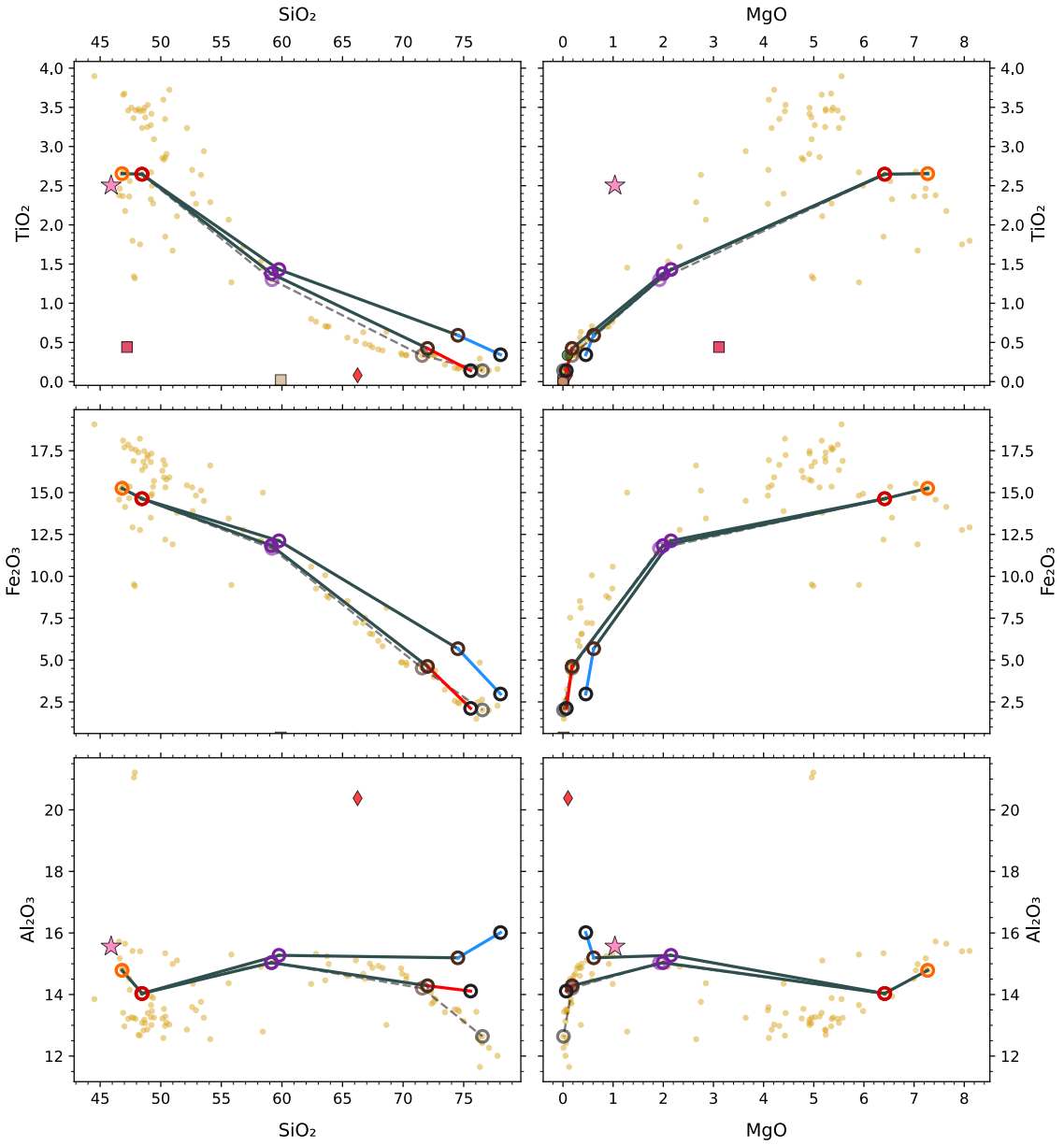
Table 7: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine	37.86	0.09	0.08	22.1	39.49	0.38	0.0	0.0
spinel								
clinopyroxene	50.67	0.0	0.96	9.29	17.28	21.49	0.24	0.07
orthopyroxene								
plagioclase	49.55	0.0	32.28	0.14	0.09	14.8	2.96	0.17
amphibole								
kfeldspar								
biotite								
quartz								
magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0
ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0

Table 8: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

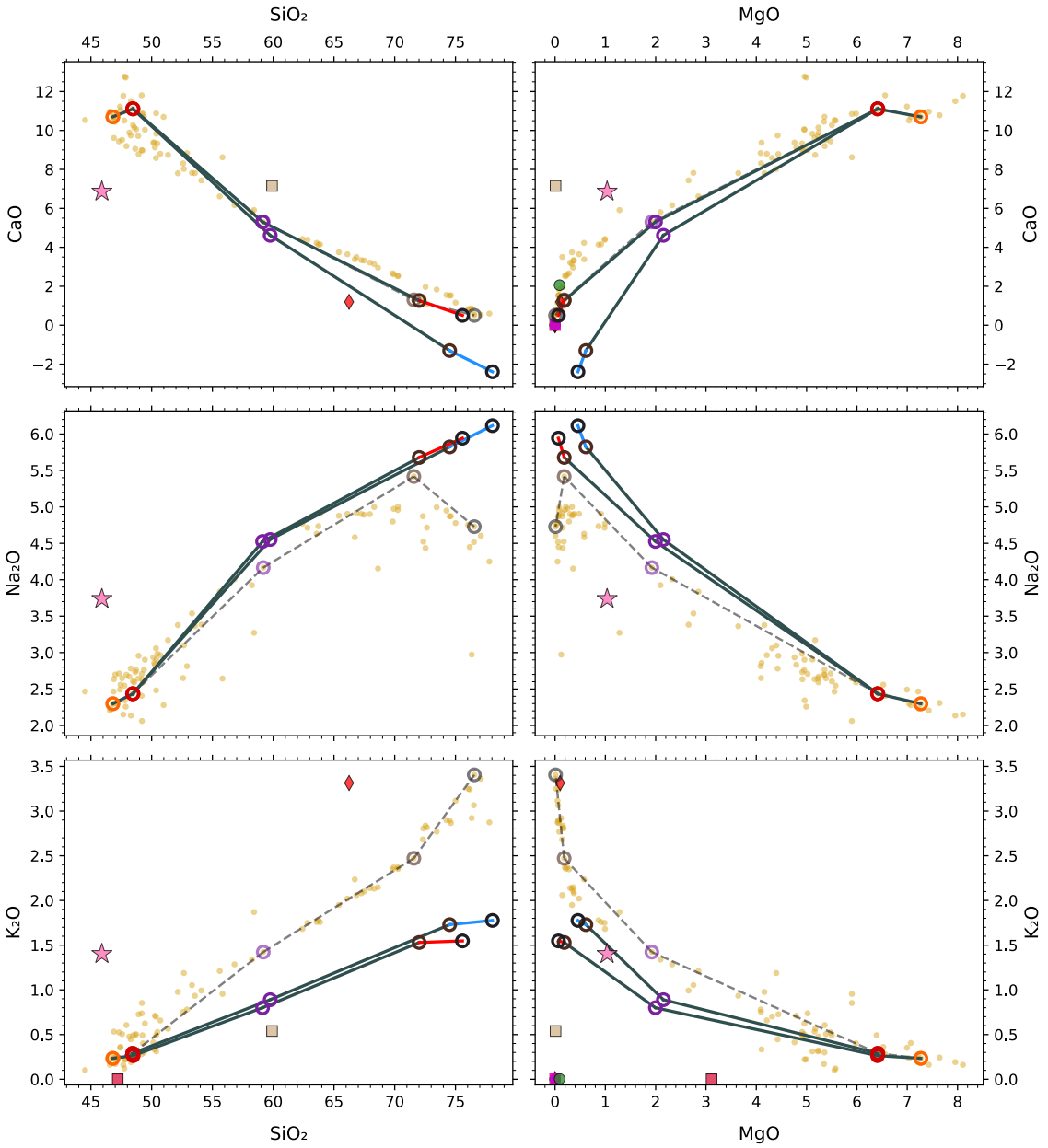


Figure 1: Bivariate plots of the data suite with the data-based and modelled LLODs. The Primitive to Basic transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Basic > Intermediate

BASIC

Population size	50
Iterations	500/500
Cost function	0.88
Trace analysis?	True

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	48.43	59.18	59.75	0.96
TiO2	2.65	1.3	1.43	10.06

	Initial	Expected	Computed	% Error
Al2O3	14.03	15.04	15.27	1.58
Fe2O3	14.63	11.67	12.12	3.89
MgO	6.41	1.92	2.15	11.89
CaO	11.12	5.3	4.62	12.95
Na2O	2.44	4.17	4.55	9.26
K2O	0.29	1.42	0.89	37.46
P	1659.08	2270.32	2092.04	7.85
Ba	108.0	390.0	317.3	18.64
Rb	13.0	39.0	40.7	4.35
Sr	230.0	0.0	258.26	258263597201991.0
Nb	15.0	56.0	44.85	19.9
Zr	133.0	645.0	414.93	35.67
Y	29.0	0.0	54.72	54717779139836.0
La	16.0	55.0	26.13	52.5
Ce	33.0	117.0	46.38	60.36
Nd	0.0	72.0		
Sm	0.0	12.5		
Eu	0.0	4.23		
Gd	0.0	15.4		
Tb	0.0	1.97		
Dy	0.0	0.0		
Ho	0.0	0.0		
Er	0.0	0.0		
Tm	0.0	1.06		
Yb	0.0	7.1		
Lu	0.0	1.0		
Hf	0.0	13.6		
Ta	0.0	3.74		
Th	0.0	5.8		
U	0.0	1.7		

Table 9: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	70.0
Olivine	7.31
Spinel	0.0
Clinopyroxene	37.72
Orthopyroxene	0.0
Plagioclase	42.39
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	7.09

	Value
Ilmenite	4.74
Apatite	0.75
Zircon	0.0
Monazite	0.0
Allanite	0.0

Table 10: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P
wt.%	43.57	3.16	13.52	15.72	8.22	13.9	1.53	0.03	1473.53

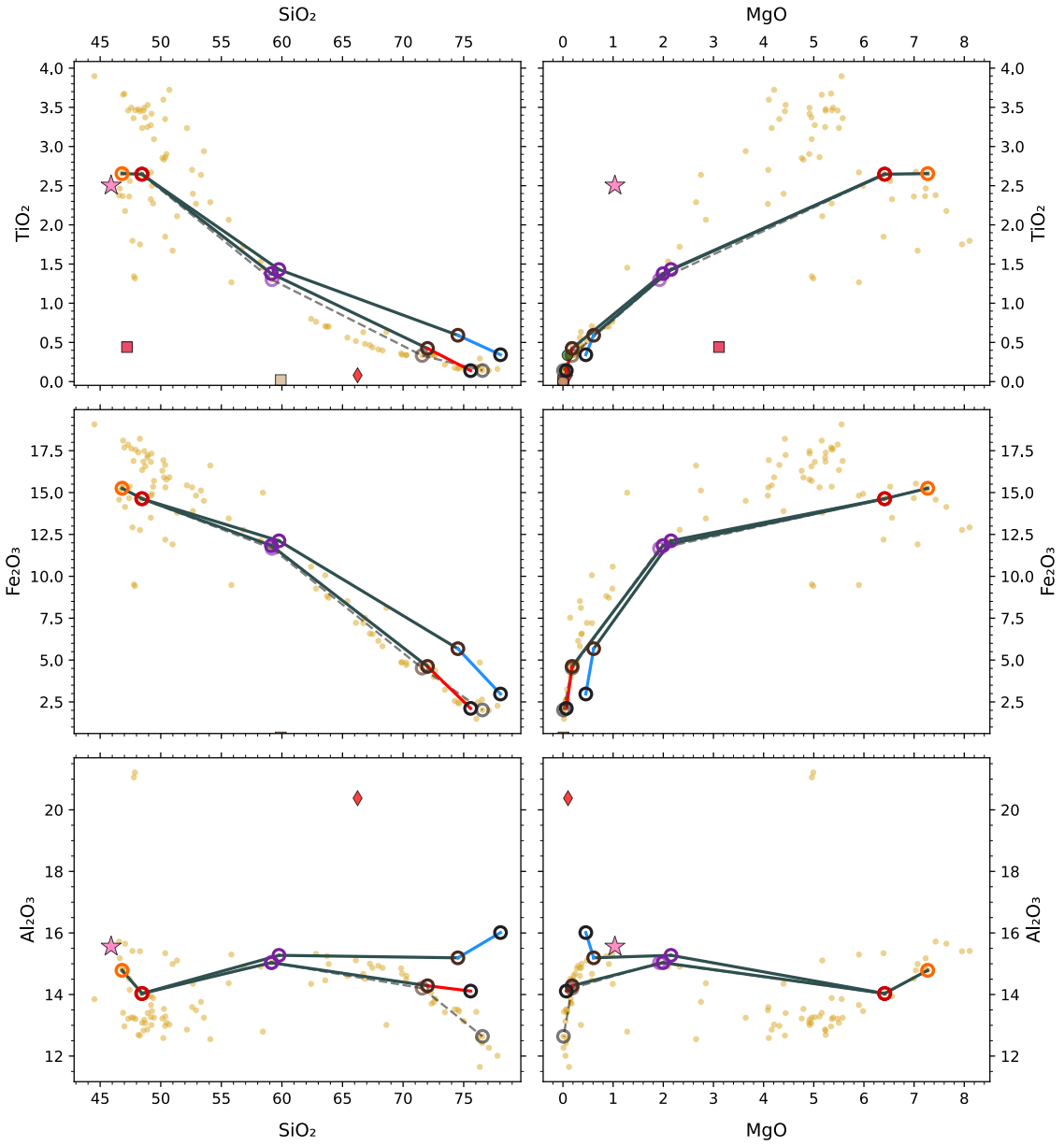
Table 11: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P
Olivine	37.59	0.04	0.41	22.62	38.91	0.42	0.0	0.0	0.0
Spinel									
Clinopyroxene	50.78	1.08	1.97	12.28	14.07	19.51	0.31	0.0	0.0
Orthopyroxene									
Plagioclase	51.11	0.07	30.07	0.78	0.17	14.39	3.33	0.08	0.0
Amphibole									
K-feldspar									
Biotite									
Quartz									
Magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0	0.0
Ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0	0.0
Apatite	0.0	0.0	0.0	0.0	0.0	55.0	0.0	0.0	196470.0
Zircon									
Monazite									
Allanite									

Table 12: Cotectic compositions of the model phases (wt.% for oxides, ppm for trace elements). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

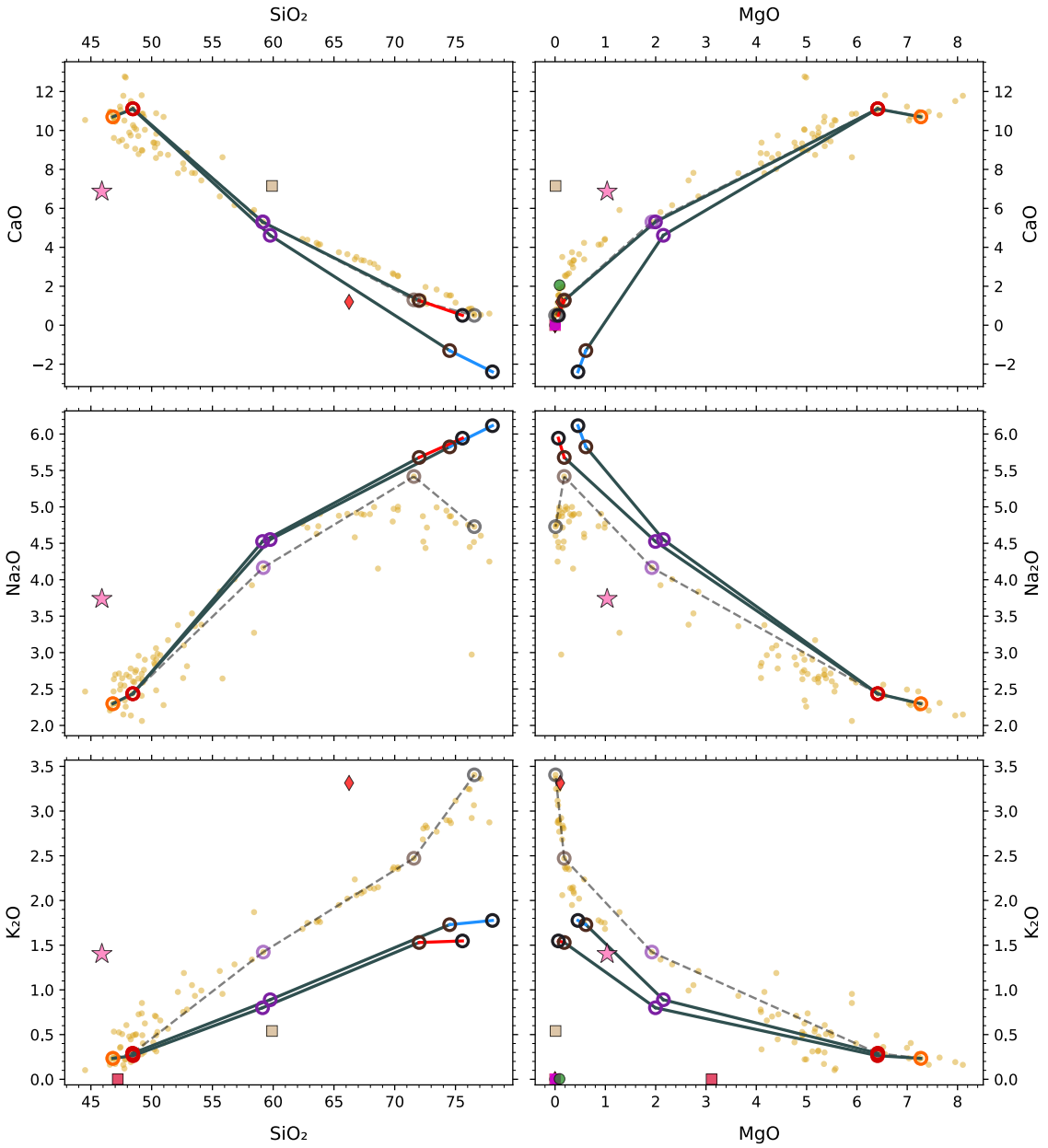


Figure 2: Bivariate plots of the data suite with the data-based and modelled LLODs. The Basic to Intermediate transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Intermediate > Evolved

INTERMEDIATE

Population size	50
Iterations	500/500
Cost function	1.02
Trace analysis?	True

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	59.75	71.57	74.52	4.12
TiO2	1.43	0.33	0.59	77.68

	Initial	Expected	Computed	% Error
Al2O3	15.27	14.2	15.19	6.96
Fe2O3	12.12	4.52	5.68	25.7
MgO	2.15	0.18	0.61	237.28
CaO	4.62	1.3	-1.3	200.05
Na2O	4.55	5.42	5.82	7.5
K2O	0.89	2.47	1.73	30.03
P	2092.04	174.64	-64.86	137.14
Ba	317.3	705.0	543.6	22.89
Rb	40.7	62.0	87.26	40.74
Sr	258.26	164.0	117.58	28.31
Nb	44.85	113.0	64.1	43.28
Zr	414.93	699.0	337.37	51.73
Y	54.72	179.0	53.7	70.0
La	26.13	99.0	33.33	66.33
Ce	46.38	180.0	44.1	75.5
Nd		0.0		
Sm		0.0		
Eu		0.0		
Gd		0.0		
Tb		0.0		
Dy		0.0		
Ho		0.0		
Er		0.0		
Tm		0.0		
Yb		0.0		
Lu		0.0		
Hf		0.0		
Ta		0.0		
Th		0.0		
U		0.0		

Table 13: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	54.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	19.45
Orthopyroxene	10.31
Plagioclase	56.27
Amphibole	0.0
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	8.83

	Value
Ilmenite	3.03
Apatite	2.0
Zircon	0.1
Monazite	0.0
Allanite	0.0

Table 14: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P	Zr
wt.%	47.17	2.15	15.34	17.61	3.47	9.66	3.47	0.17	3929.4	481.0

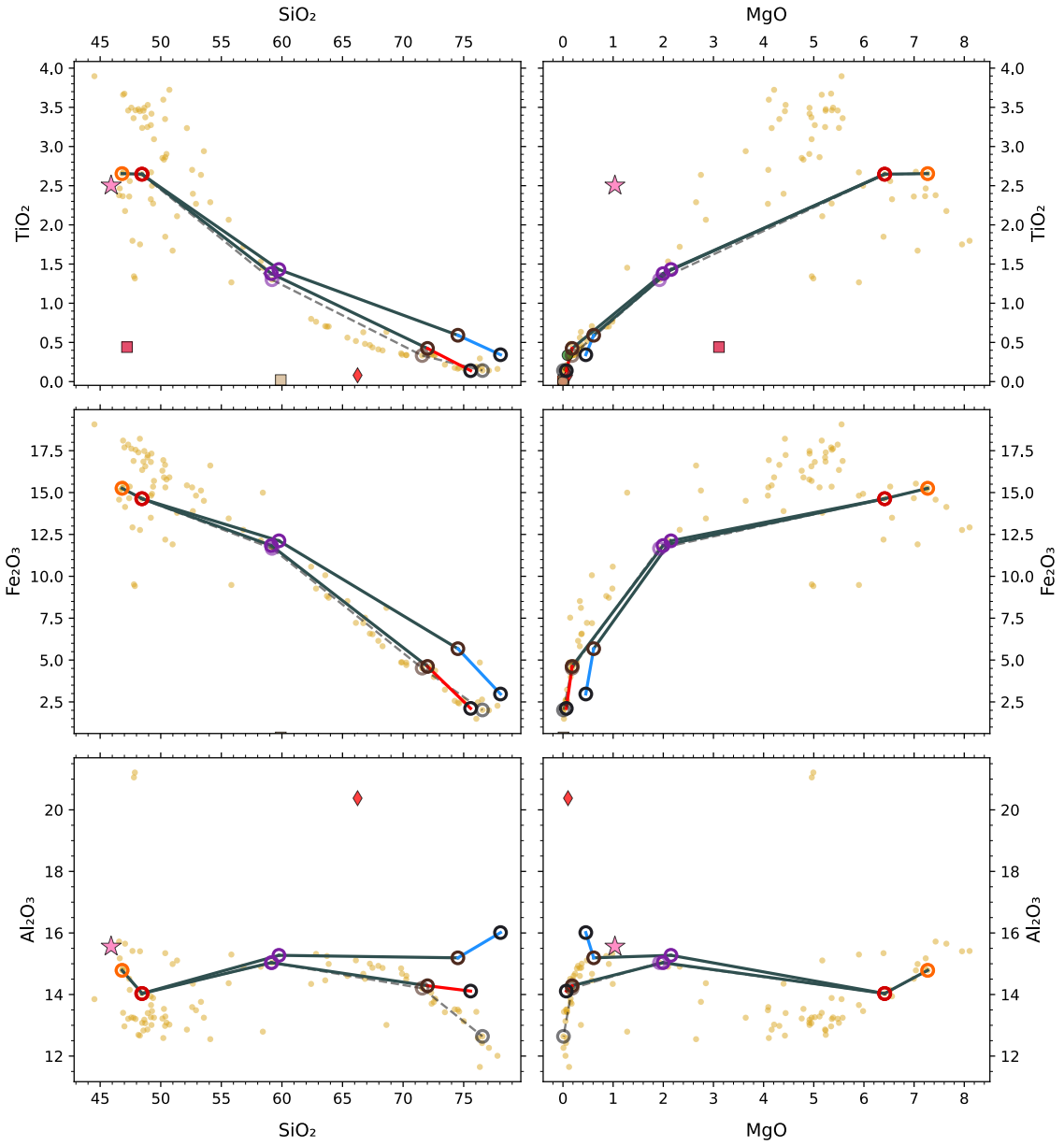
Table 15: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P	Zr
Olivine										
Spinel										
Clinopyroxene	48.86	0.76	1.65	22.42	9.21	16.86	0.24	0.0	0.0	0.0
Orthopyroxene	49.37	0.16	3.07	29.6	16.07	1.73	0.0	0.0	0.0	0.0
Plagioclase	57.84	0.04	26.13	0.51	0.03	9.06	6.08	0.31	0.0	0.0
Amphibole										
K-feldspar										
Biotite										
Quartz										
Magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0	0.0	0.0
Ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0	0.0	0.0
Apatite	0.0	0.0	0.0	0.0	0.0	55.0	0.0	0.0	196470.0	0.0
Zircon	35.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	481000.0
Monazite										
Allanite										

Table 16: Cotectic compositions of the model phases (wt.% for oxides, ppm for trace elements). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

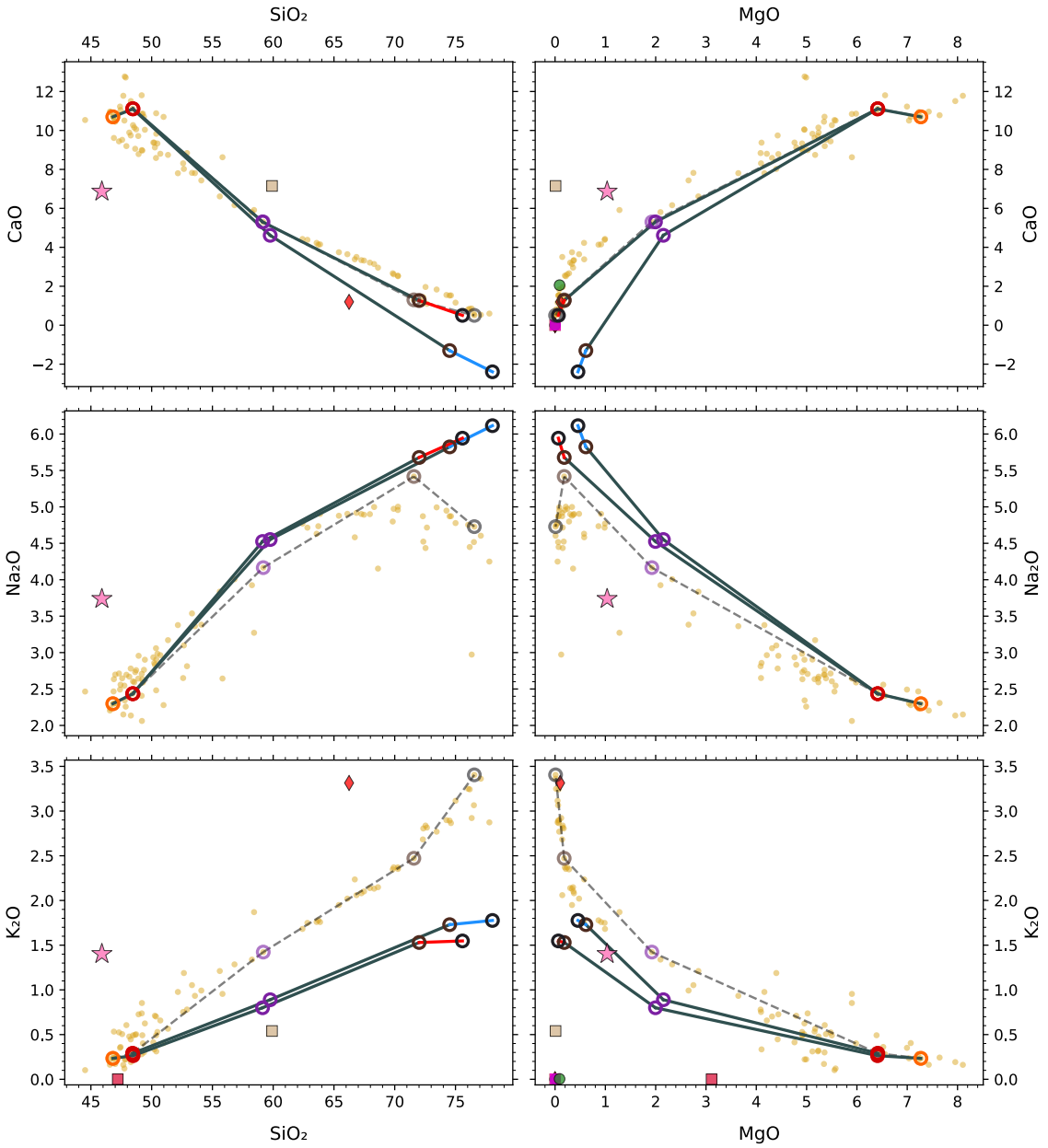


Figure 3: Bivariate plots of the data suite with the data-based and modelled LLODs. The Intermediate to Evolved transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Evolved > End

EVOLVED

Population size	50
Iterations	500/500
Cost function	2.72
Trace analysis?	True

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	74.52	76.55	78.04	1.95
TiO2	0.59	0.14	0.34	142.44

	Initial	Expected	Computed	% Error
Al2O3	15.19	12.64	16.01	26.68
Fe2O3	5.68	2.02	2.97	47.1
MgO	0.61	0.01	0.46	4405.64
CaO	-1.3	0.51	-2.39	573.4
Na2O	5.82	4.73	6.11	29.31
K2O	1.73	3.41	1.78	47.83
P	-64.86	43.66	-274.64	729.03
Ba	543.6	834.0	286.16	65.69
Rb	87.26	103.0	96.83	5.99
Sr	117.58	57.0	90.7	59.13
Nb	64.1	131.0	52.06	60.26
Zr	337.37	329.0	55.42	83.15
Y	53.7	194.0	50.1	74.17
La	33.33	136.0	34.99	74.27
Ce	44.1	243.0	44.94	81.51
Nd		0.0		
Sm		0.0		
Eu		0.0		
Gd		0.0		
Tb		0.0		
Dy		0.0		
Ho		0.0		
Er		0.0		
Tm		0.0		
Yb		0.0		
Lu		0.0		
Hf		0.0		
Ta		0.0		
Th		0.0		
U		0.0		

Table 17: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	12.0
Olivine	20.56
Spinel	0.0
Clinopyroxene	30.93
Orthopyroxene	0.0
Plagioclase	1.05
Amphibole	0.0
K-feldspar	41.59
Biotite	0.0
Quartz	0.02
Magnetite	0.16

	Value
Ilmenite	4.45
Apatite	0.75
Zircon	0.5
Monazite	0.0
Allanite	0.0

Table 18: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P	Zr
wt.%	48.66	2.41	9.24	25.61	1.72	6.63	3.69	1.38	1473.53	2405.0

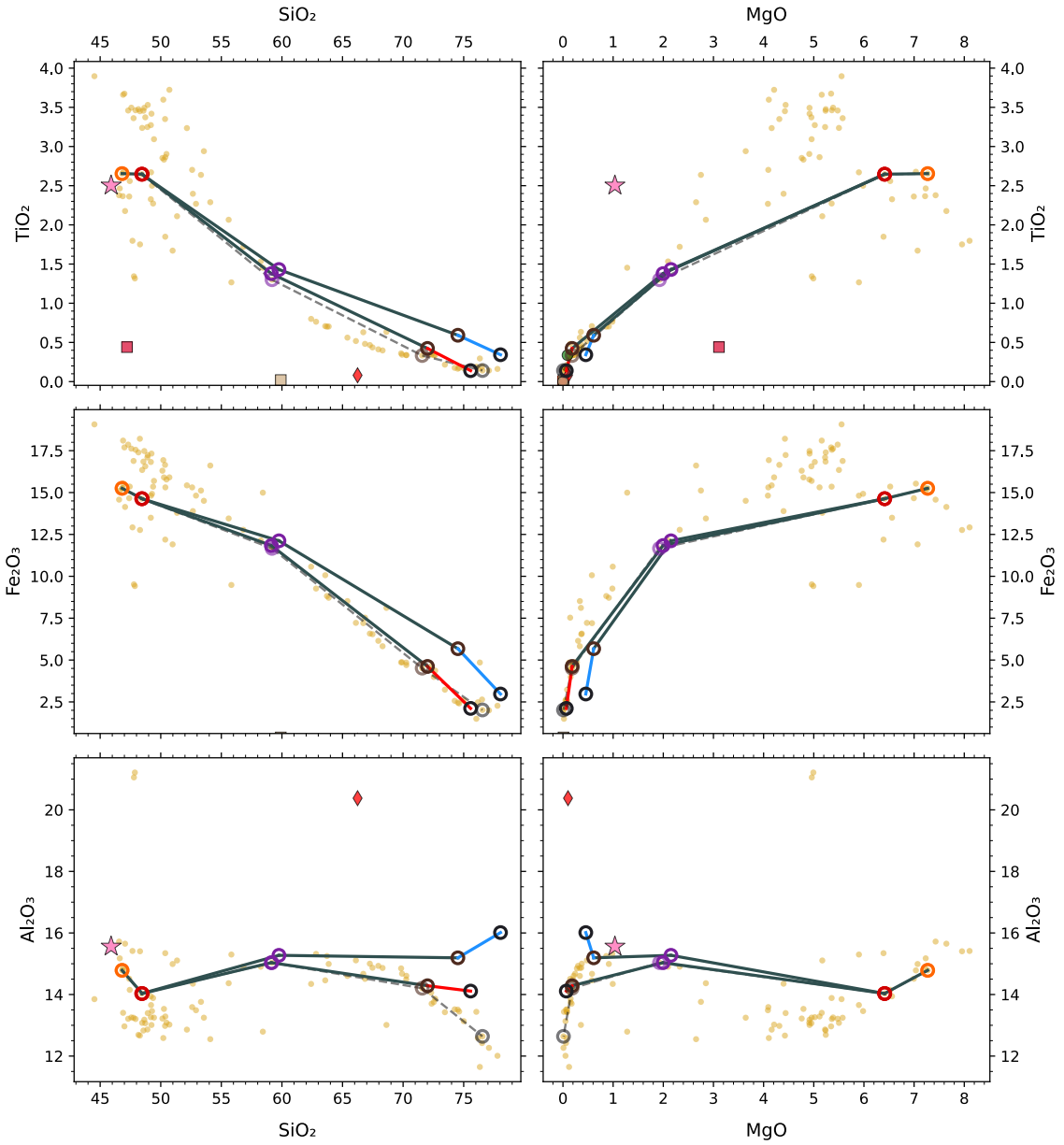
Table 19: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O	P	Zr
Olivine	27.67	0.05	1.01	68.46	3.47	-0.66	0.0	0.0	0.0	0.0
Spinel										
Clinopyroxene	47.21	0.44	0.95	29.34	3.11	18.7	0.25	0.0	0.0	0.0
Orthopyroxene										
Plagioclase	59.91	0.02	24.85	0.38	0.01	7.15	7.14	0.54	0.0	0.0
Amphibole										
K-feldspar	66.24	0.08	20.38	0.18	0.1	1.2	8.5	3.31	0.0	0.0
Biotite										
Quartz	100.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0	0.0	0.0
Ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0	0.0	0.0
Apatite	0.0	0.0	0.0	0.0	0.0	55.0	0.0	0.0	196470.0	0.0
Zircon	35.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	481000.0
Monazite										
Allanite										

Table 20: Cotectic compositions of the model phases (wt.% for oxides, ppm for trace elements). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

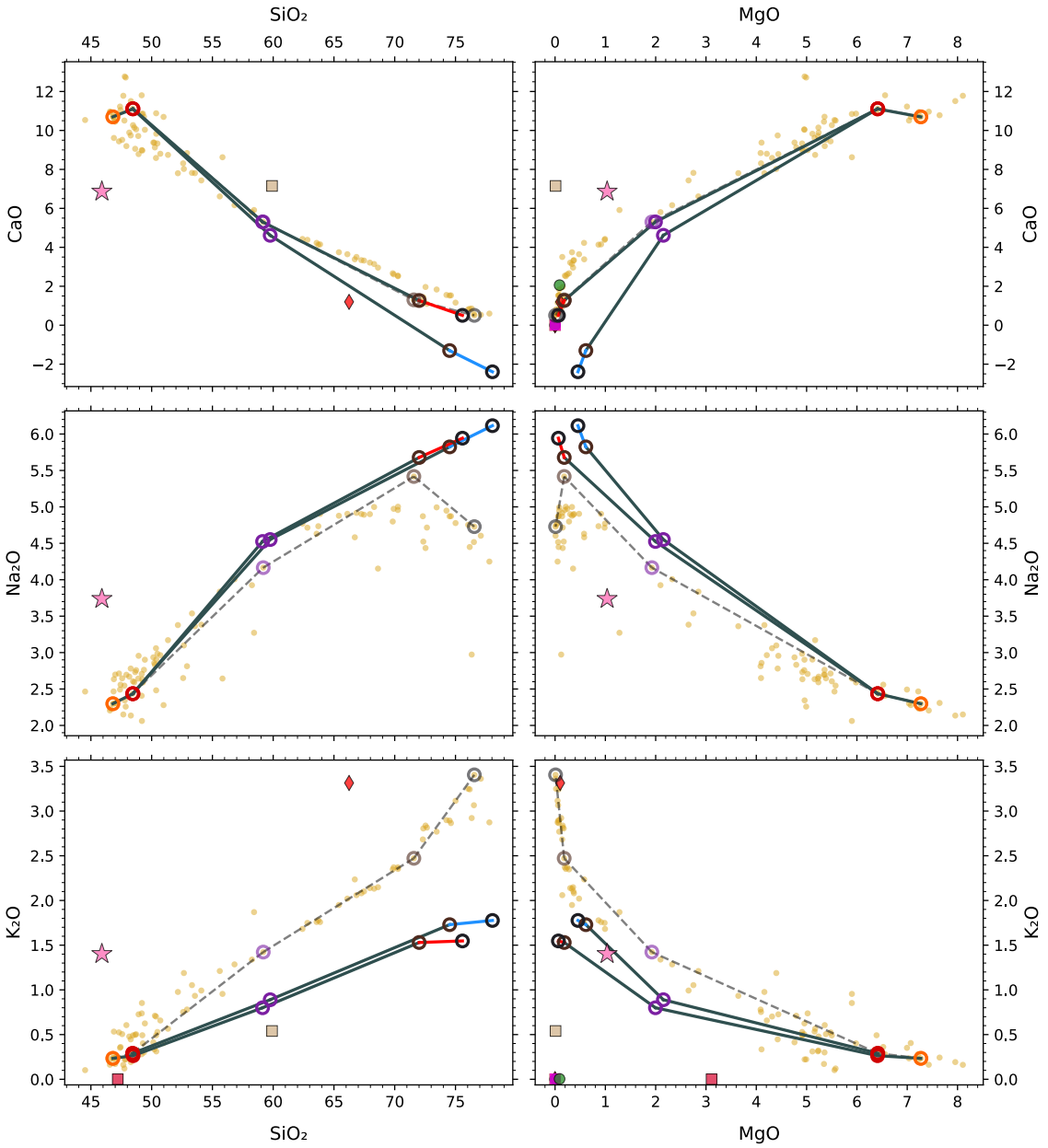


Figure 4: Bivariate plots of the data suite with the data-based and modelled LLODs. The Evolved to End transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

5. Figures

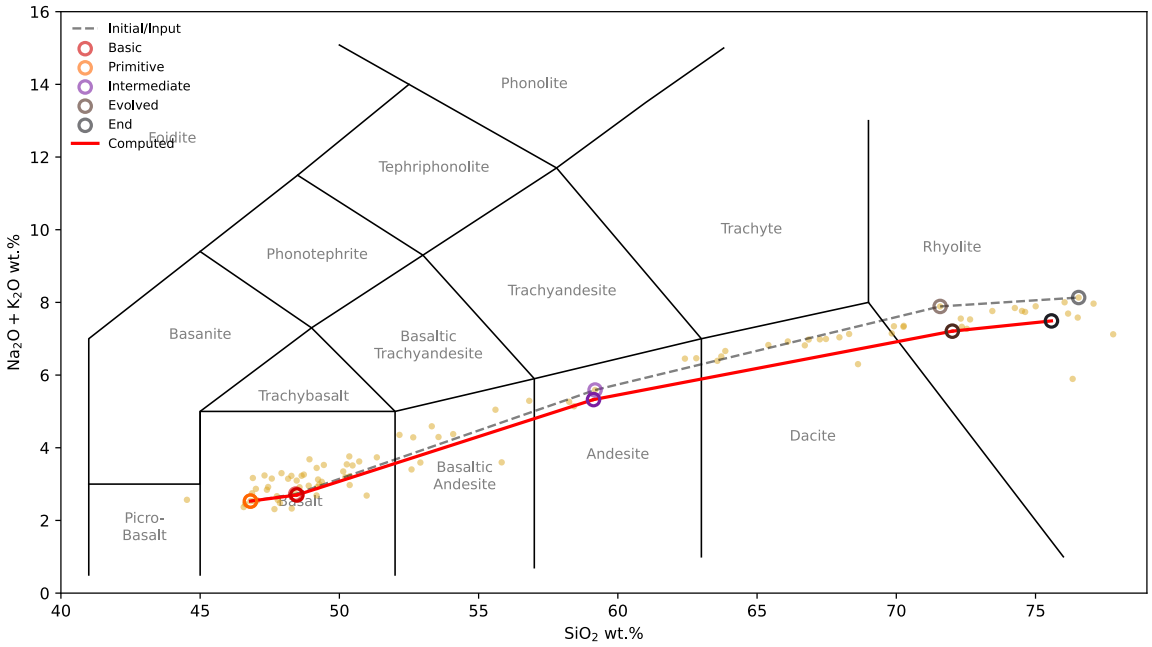


Figure 5: Total Alkali-Silica (TAS) diagram of the model phases.

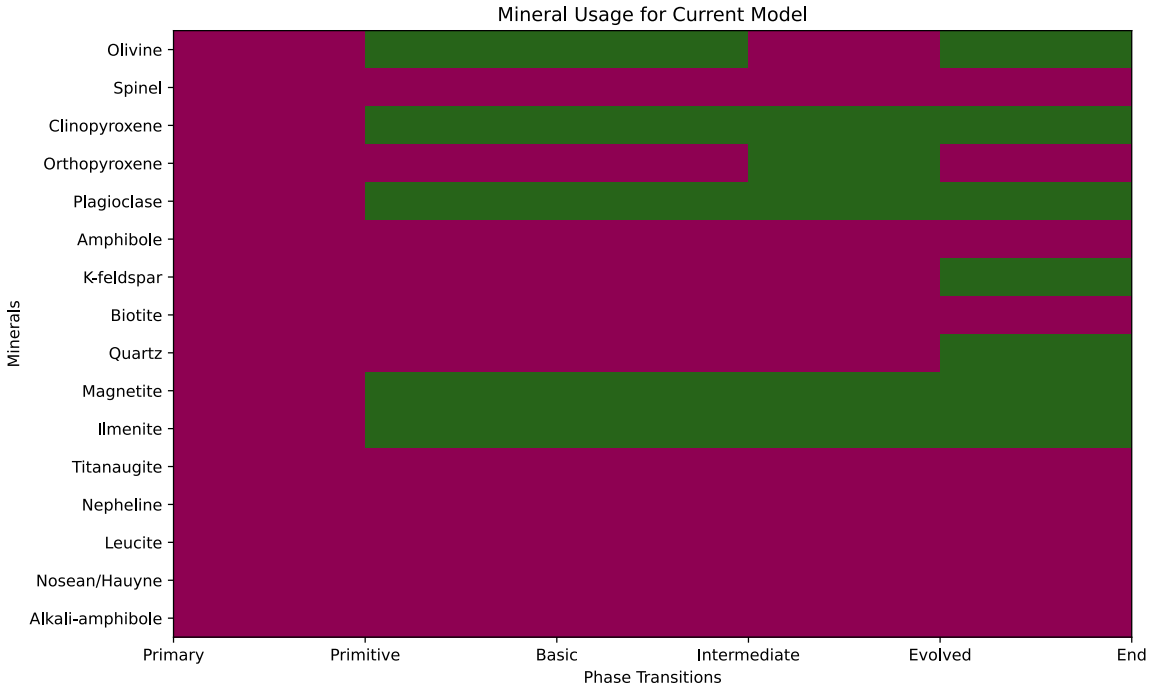


Figure 6: Mineral usage across the model phases.

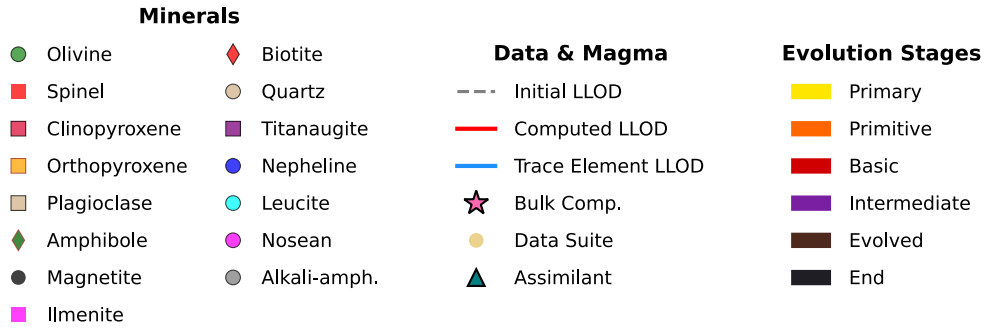


Figure 7: Legend for mineral symbols used in the figures.